# Jiamin Zhou, MSc

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## Employment History

2025 – · · · · **Staff Research Associate IV** – University of California, San Francisco
SRA IV in Dr. Aaron Field's Laboratory for Orthopaedic Biomechanics and Biotransport in the Department of Orthopaedic Surgery
Currently leading a new study on the development and validation of age-relative measures of biochemical intervertebral disc health (NIH R01AR085550)
Advanced MRI biomarker analysis, meta-genome-wide association study (GWAS) analysis for developing polygenic risk score, management of clinical research study data assets and databases

2019 – 2025 **Staff Research Associate III** – University of California, San Francisco
SRA III in Dr. Aaron Field's lab (2022–2025) and Dr. Roland Krug's Musculoskeletal Magnetic Resonance Imaging Lab in the Department of Radiology and Biomedical Imaging (2019–2022)
Characterization of chronic low back pain (Back Pain Consortium Research Program) through automatic vertebral body and paraspinal muscle segmentation, fat quantification of whole-body MRI, statistical analysis of multi-omic data, processing of whole genome sequencing data.

2016 – 2017 **Summer Undergraduate Research Fellowship** – University of Texas Southwestern Medical Center
Research fellow in Dr. Qinghua Liu's lab in the Department of Biochemistry
Studied the molecular mechanism of odor-induced innate fear mediated by Trpa1 in mice. Dissected and sectioned mouse brain tissue, modified a PCR protocol to target and produce a restriction enzyme recognition site on Trpa1, developed a method to selectively label bound proteins using biotinylation.

2013 – 2014 **Research Intern** – University of Texas Southwestern Medical Center
Intern in Dr. Zhigao Wang's lab in the Department of Molecular Biology
Studied how the herb *Citrullus colocynthis* kills cancer cells through necrosis. Developed a technique to detect phosphorylated proteins in Western blots.

## Education

2018 – 2019 **MSc in Biomedical Imaging** – University of California, San Francisco
Thesis title: *Spectrome-AI: a Neural Network Framework for Inferring MEG Spectra* (supervised by Dr. Ashish Raj)
Experience with diffusion tensor imaging acquisition and reconstruction of tractograms, advanced MR acquisitions and techniques, correction of motion artifacts and reconstruction of higher field MR images using CNNs

2015 – 2018 **BA (Hons) in Natural Sciences** – University of Cambridge
Completed a final year dissertation on modelling calcium feedback mechanisms in light adaptation in vertebrate cone photoreceptors using XPPAUT to solve coupled first order ODEs (supervised by Dr. Hugh Matthews)

## Teaching Experience

2020 – 2022 **Bootcamp Instructor** – University of California, San Francisco
MSBI Bootcamp (Fall 2020, Fall 2021, Fall 2022)
Taught incoming Master's students with no prior programming background how to code in MATLAB

2019 – 2020 **Teaching Assistant** – University of California, San Francisco
MR Pulse Sequences (Spring 2020)
Physical Principles of CT, PET, and SPECT Imaging (Fall 2019)

## Awards

2026 **ISSLS Prize in Clinical Science** – International Society for the Study of the Lumbar Spine
Co-author of 'Data-driven classification of lumbar spine degeneration trajectories in chronic low back pain'

2023 **ISSLS Prize in Bioengineering Science** – International Society for the Study of the Lumbar Spine
Co-author of 'Age- and sex-related differences in lumbar intervertebral disc degeneration between patients with chronic low back pain and asymptomatic controls'

## Talks

2025 **International Society for the Study of the Lumbar Spine** – Atlanta, GA, USA
'Serum inflammatory profiles associate with intradiscal propionic acid levels in chronic low back pain patients with Modic changes' (oral presentation, shortlisted for Best Paper Award)
'Modic changes associate with gut microbial dysbiosis' (oral presentation)

**BACPAC Spring Meeting** – Chapel Hil, NC, USA
'Phenotyping of comeBACK patients with Modic changes using MRI, MRS, and serum cytokines' (oral presentation)

2024 **International Society for the Study of the Lumbar Spine** – Milan, Italy
'Magnetic resonance spectroscopy biomarker of Modic changes with bacterial etiology' (oral presentation, shortlisted for Best Paper Award)

## Skills

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|---|---|
| Research | Experimental design, MRI physics, statistical data analysis, statistical modeling, technical writing, clinical database management, REDCap, ELISA, PCR |
| Programming | Python, numpy, scipy, scikit-learn, Tensorflow, matplotlib, seaborn, plotly, pandas, Jupyter, MATLAB, R, LaTeX, Java, PHP, SQL |
| Tools | git version control, bash, HPC and job scheduling in SLURM/SGE, JMP, Nextflow, nf-core, MS Office, Notion |
| Languages | English, Mandarin Chinese (speaking). |

## Research Publications

### Preprints


1. J. Devan, **J. Zhou**, P.-H. Wu, N. Bonnheim, C. O'Neill, J. C. Lotz, T. M. Link, A. Torres-Espin, S. Dudli, A. J. Fields, and t. R. C. at Ucsf, *Intradiscal propionic acid levels in chronic low back pain patients with modic type 1 changes are associated with systemic immune cell activation*, Jan. 2026. DOI: `10.64898/2026.01.06.26343445`

2. J. Devan, T. Mengis, P. Bitterli, **J. Zhou**, P. Leary, P.-H. Wu, T. Link, O. Distler, A. Fields, S. Dudli, and t. R. Investigators, *HLA-A*26:01 is associated with vertebral endplate bone marrow lesions (Modic changes) in chronic low back pain: A novel immune-genetic link*, Sep. 2025. DOI: `10.1101/2025.09.29.25336889`


### Journal Articles


1. T. P. McSweeney, Z. Akkaya, **J. Zhou**, P.-H. Wu, N. B. Bonnheim, T. M. Link, J. Karppinen, J. C. Lotz, S. Saarakkala, A. J. Fields, and A. Tiulpin, "Data-driven classification of lumbar spine degeneration trajectories in chronic low back pain," *European Spine Journal*, Feb. 2026, ISSN: 1432-0932. DOI: `10.1007/s00586-026-09840-1`

2. K. Khattab, L. K. Dziesinski, J. Ornowski, **J. Zhou**, N. B. Bonnheim, R. Crawford, A. Scheffler, A. J. Fields, C. W. O'Neill, J. C. Lotz, and J. F. Bailey, "Spatial patterns of fat within the deep multifidus as a biomarker for chronic low back pain," English, *The Spine Journal*, vol. 0, no. 0, Jul. 2025, ISSN: 1529-9430, 1878-1632. DOI: `10.1016/j.spinee.2025.07.022`

3. K. Khattab, L. K. Dziesinski, **J. Zhou**, A. Scheffler, A. J. Fields, C. W. O'Neill, J. C. Lotz, and J. F. Bailey, "Chronicity and physical activity levels independently associate with spatial patterns of multifidus fat infiltration in chronic low back pain patients," *European Spine Journal*, Dec. 2025, ISSN: 1432-0932. DOI: `10.1007/s00586-025-09660-9`

4. M. W. Tong, **J. Zhou**, Z. Akkaya, S. Majumdar, and R. Bhattacharjee, "Artificial intelligence in musculoskeletal applications: A primer for radiologists," *Diagnostic and Interventional Radiology*, Mar. 2025, ISSN: 1305-3612. DOI: `10.4274/dir.2024.242830`

5. N. B. Bonnheim, A. A. Lazar, A. Kumar, Z. Akkaya, **J. Zhou**, X. Guo, C. O'Neill, T. M. Link, J. C. Lotz, R. Krug, and A. J. Fields, "Issls prize in bioengineering science 2023: Age- and sex-related differences in lumbar intervertebral disc degeneration between patients with chronic low back pain and asymptomatic controls," en, *European Spine Journal*, vol. 32, no. 5, pp. 1517–1524, May 2023, ISSN: 1432-0932. DOI: `10.1007/s00586-023-07542-6`

6. N. B. Bonnheim, L. Wang, A. A. Lazar, R. Chachad, **J. Zhou**, X. Guo, C. O'Neill, J. Castellanos, J. Du, H. Jang, R. Krug, and A. J. Fields, "Deep-learning-based biomarker of spinal cartilage endplate health using ultra-short echo time magnetic resonance imaging," en, *Quantitative Imaging in Medicine and Surgery*, vol. 13, no. 5, pp. 2 807 821–2 802 821, May 2023, ISSN: 2223-4306, 2223-4292. DOI: `10.21037/qims-22-729`

7. N. B. Bonnheim, L. Wang, A. A. Lazar, **J. Zhou**, R. Chachad, N. Sollmann, X. Guo, C. Iriondo, C. O'Neill, J. C. Lotz, T. M. Link, R. Krug, and A. J. Fields, "The contributions of cartilage endplate composition and vertebral bone marrow fat to intervertebral disc degeneration in patients with chronic low back pain," en, *European Spine Journal*, vol. 31, no. 7, pp. 1866–1872, Jul. 2022, ISSN: 1432-0932. DOI: `10.1007/s00586-022-07206-x`

8. N. Sollmann, N. B. Bonnheim, G. B. Joseph, R. Chachad, **J. Zhou**, Z. Akkaya, A. M. Pirmoazen, J. F. Bailey, X. Guo, A. A. Lazar, T. M. Link, A. J. Fields, and R. Krug, "Paraspinal muscle in chronic low back pain: Comparison between standard parameters and chemical shift encoding-based water–fat mri," en, *Journal of Magnetic Resonance Imaging*, vol. 56, no. 5, pp. 1600–1608, 2022, ISSN: 1522-2586. DOI: `10.1002/jmri.28145`

9. **J. Zhou**, P. F. Damasceno, R. Chachad, J. R. Cheung, A. Ballatori, J. C. Lotz, A. A. Lazar, T. M. Link, A. J. Fields, and R. Krug, "Automatic vertebral body segmentation based on deep learning of dixon images for bone marrow fat fraction quantification," English, *Frontiers in Endocrinology*, vol. 11, Sep. 2020, ISSN: 1664-2392. DOI: `10.3389/fendo.2020.00612`

10. X. Wang, **J. Zhou**, C. Qi, and G. Wang, "Establishing cell lines overexpressing dr3 to assess the apoptotic response to anti-mitotic therapeutics," en, *Journal of Visualized Experiments (JoVE)*, no. 143, e58705, Jan. 2019, ISSN: 1940-087X. DOI: `10.3791/58705`

11. Y. Wang, L. Cao, C.-Y. Lee, T. Matsuo, K. Wu, G. Asher, L. Tang, T. Saitoh, J. Russell, D. Klewe-Nebenius, L. Wang, S. Soya, E. Hasegawa, Y. Chérasse, **J. Zhou**, Y. Li, T. Wang, X. Zhan, C. Miyoshi, Y. Irukayama, J. Cao, J. P. Meeks, L. Gautron, Z. Wang, K. Sakurai, H. Funato, T. Sakurai, M. Yanagisawa, H. Nagase, R. Kobayakawa, K. Kobayakawa, B. Beutler, and Q. Liu, "Large-scale forward genetics screening identifies trpa1 as a chemosensor for predator odor-evoked innate fear behaviors," en, *Nature Communications*, vol. 9, no. 1, p. 2041, May 2018, ISSN: 2041-1723. DOI: `10.1038/s41467-018-04324-3`

## References

Available upon request.